

Warn the operator before permitted headroom runs out

We look at how much the generators have already run, what tests are coming and what events may force them to run — and warn the operator before they run out of permitted headroom.



Know what has run.

Plan what must run.

Preserve headroom for what may happen.

For data-centre compliance and critical-facilities teams deciding which generators can run, when and for how long.

A four-hour request is not always a four-hour decision

REQUEST

Run 20 standby generators for a four-hour event next Tuesday.

THE DECISION

Can we proceed?

If yes: which generators, for how long, and what must remain available afterwards?

BEFORE APPROVING IT, THE TEAM MUST RECONCILE

- 8 generators were recently tested
- 2 run records are incomplete
- 1 generator has an open maintenance issue
- ✦ more mandatory tests are already scheduled
- ! severe weather may increase emergency need

Today, the answer is assembled across permits, spreadsheets, controllers, maintenance systems and operator logs.

A clear recommendation — with the evidence behind it

PROCEED WITH CONDITIONS

Use Generators 1–8 for no more than two hours.

Do not use Generators 9–10

Exclude Generator 14

Why two hours?

Previous runs are not yet classified

Open maintenance condition

It leaves enough permitted headroom for next month’s tests and the selected eight-hour outage case.

WHAT THE OPERATOR CAN SEE	
Now	30 h
After tests	18 h
After event	12 h
8-hour outage	–8 h

If evidence is missing
Say “cannot determine” and
name what is needed.

The product recommends and explains. The accountable team decides and operates the generators.

Make the decision before the generators run

TODAY

Teams prove what happened afterwards.

- Search permits, logs and spreadsheets
- Find missing records during review
- Schedule tests one at a time
- Guess what must remain for emergencies
- Treat grid opportunities cautiously

WITH THE PLANNER

Teams test the plan before approving it.

- ✓ Spend less time assembling the answer
- ✓ Find evidence gaps before approval
- ✓ Plan tests against the same headroom
- ✓ See what remains under agreed outage cases
- ✓ Assess them with visible constraints

The practical value: less manual effort, earlier warnings, better test planning and more confidence in what can safely run.

The data is familiar — the challenge is joining it

1	What is allowed?	Permits + amendments
2	What has run?	Controllers / EPMS
3	Why did it run?	Operator + incident logs
4	What must run next?	Test schedule / CMMS
5	What is unavailable?	Maintenance records
6	What may force a run?	Weather + grid signals

HOW THE ANSWER IS BUILT

- 1 Read permit rules and match records to each generator
- 2 Work out what ran, why it ran and what is still uncertain
- 3 Calculate remaining headroom and test the proposed plan
- 4 Give the operator a recommendation with evidence

AI reads and connects the evidence. Approved rules perform the calculation. Qualified people make the decision.

What must be true before a pilot starts

The concept should be tested against a real site and a known historical decision before any full build.

✓	Current permit	Including amendments and accepted interpretations
✓	Verified generator map	Permit names, asset IDs and controller tags agree
✓	Credible run history	Runtime, starts and load are sufficiently complete
✓	Purpose evidence	Most activity can be classified or marked unresolved
✓	Named reviewer	A compliance owner can validate rules and exceptions

If these conditions are not met, stop, narrow the scope or reposition the use case.

Plain language on the left; technical capability on the right

WHAT THE CLIENT HEARS	WHAT WE BUILD	WHAT IT DOES
 Read permit rules	Document intelligence	Extract limits, exceptions, dates and citations
 Match records to generators	Entity resolution	Link permit names, asset IDs, tags and work orders
 Work out why each run happened	Event reconstruction	Combine logs, outages, work orders and schedules
 Calculate permitted headroom	Rules + arithmetic	Use approved compliance logic and calculation versions
 Test different outage cases	Scenario analysis	Show what remains under explicit 2h, 8h and 24h cases
 Find missing evidence	Agentic workflow	Request evidence, route review and assemble the pack

Control boundary: read-only recommendation; no self-approval, legal opinion or autonomous generator control.